

Hadoop Practical No 3 – Student Grade MapReduce

STEP 1 — Go to Home Directory

```
cd ~
```

STEP 2 — Create Project Folder

```
mkdir ~/studentgrades
```

```
cd ~/studentgrades
```

STEP 3 — Create Mapper File

```
nano GradeMapper.java
```

GradeMapper.java

```
import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Mapper;

public class GradeMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        String[] fields = line.split(",");

        String student = fields[0];

        int marks = Integer.parseInt(fields[1]);

        context.write(new Text(student), new IntWritable(marks));
    }
}
```

STEP 4 — Create Reducer File

```
nano GradeReducer.java
```

```
import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Reducer;

public class GradeReducer extends Reducer<Text, IntWritable, Text, Text> {

    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int total = 0;

        for (IntWritable val : values) {
            total += val.get();
        }

        String grade;

        if (total >= 90) {
            grade = "A";
        } else if (total >= 80) {
            grade = "B";
        } else if (total >= 70) {
            grade = "C";
        } else if (total >= 60) {
            grade = "D";
        } else {
            grade = "F";
        }

        context.write(key, new Text(grade));
    }
}
```

STEP 5 — Create Driver File

nano GradeDriver.java

```
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GradeDriver {

    public static void main(String[] args) throws Exception {

        Configuration conf = new Configuration();

        Job job = Job.getInstance(conf, "Student Grades");

        job.setJarByClass(GradeDriver.class);

        job.setMapperClass(GradeMapper.class);

        job.setReducerClass(GradeReducer.class);

        job.setOutputKeyClass(Text.class);

        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));

        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);

    }
}
```

STEP 6 — Compile Java Files

javac -classpath `hadoop classpath` -d . GradeMapper.java GradeReducer.java GradeDriver.java

STEP 7 — Create JAR File

jar -cvf grades.jar *.class

STEP 8 — Create Input File

nano grades.txt

```
Rahul,95
Amit,82
Sneha,76
Priya,65
Rohan,45
```

STEP 9 — Upload File to HDFS

```
hdfs dfs -mkdir /gradesinput
hdfs dfs -put grades.txt /gradesinput
hdfs dfs -ls /gradesinput
```

STEP 10 — Run MapReduce Program

hadoop jar grades.jar GradeDriver /gradesinput /gradesoutput

STEP 11 — View Output

hdfs dfs -cat /gradesoutput/part-r-00000

Expected Output:

```
Amit B
Priya D
Rahul A
```

Rohan F
Sneha C

STEP 12 — Before Running Again

```
hdfs dfs -rm -r /gradesoutput
```

Important Viva Points

- Mapper reads student name and marks
- Reducer calculates grade
- Driver connects Mapper and Reducer